How offline tracking looks like for Phase-2?

Breno Orzari
Centre de Calcul de l'IN2P3 / CNRS
On behalf of the TRK POG

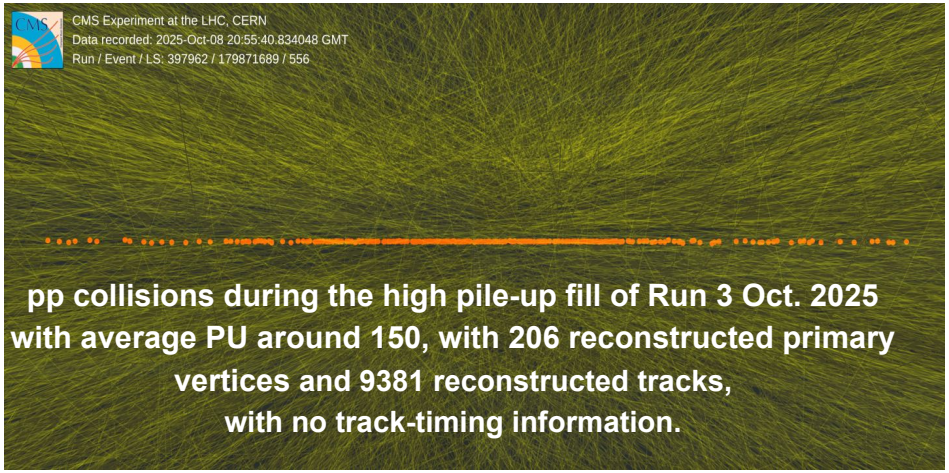
email: cms-pog-conveners-tracking@cern.ch

Tracker DPG Workshop
https://indico.cern.ch/event/1707530/

VUB/ULB - Brussels, 16th October 2026

Phase-2 Tracking Challenges and Targets

- High Luminosity LHC (HL-LHC):
  - New physics opportunities
  - New challenges
- Increased luminosity
  - High number of PU: Up to 200 (Run 5)
  - Large number of tracks from charged particles $O(10^3)$ in Run 3 ➡ $O(10^4)$ in HL-LHC
  - Superlinear increase of computational complexity
    - Due to the combinatorial nature of track reconstruction
- This leads to:
  - Increased timing: can the algorithms keep up w/ the data rate?
  - Increased cost: more processing power ⇒ Larger budget…
- By CSA28, need to work towards:
  - Minimize timing
  - Retain/extend physics performance
  - Most of developments are already merged in CMSSW, under refinement stage, expect to be in very good shape for Spring 2027



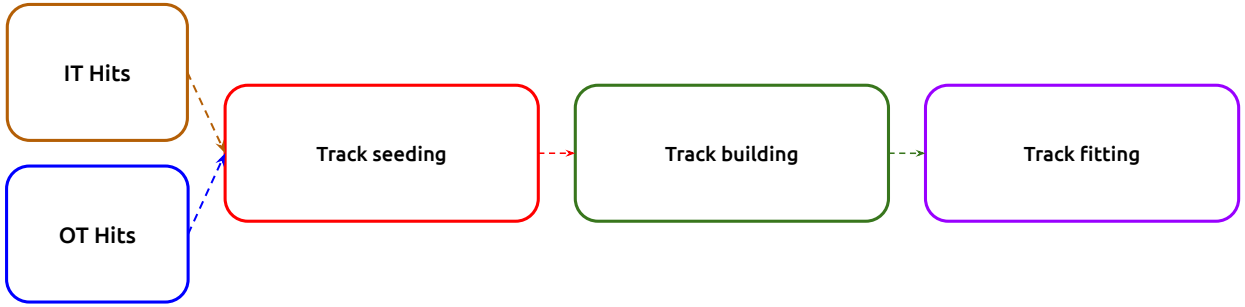
pp collisions during the high pile-up fill of Run 3 Oct. 2025 with average PU around 150, with 206 reconstructed primary vertices and 9381 reconstructed tracks, with no track-timing information.

Our Goals

- Fully exploit computing capabilities
  - Vectorized and parallelized algorithms
  - Heterogeneous computing
- Combine computing paradigm shift with physics performance improvements (e.g. ML methods) to:
  - Improve efficiency of already covered phase space
  - Extend phase space probed, e.g. displaced tracks

Tracking (at the HL-LHC)

- Traditionally, track reconstruction follows three main steps:
  1. Track **seeding**: seeds are made from hits on subset of tracker layers
  2. Track **building**: seeds are extended using full tracker information
  3. Track **fitting**: built tracks are fitted, and track parameters extracted
  - ★ Track **selection**: optionally, final tracks are selected based on quality

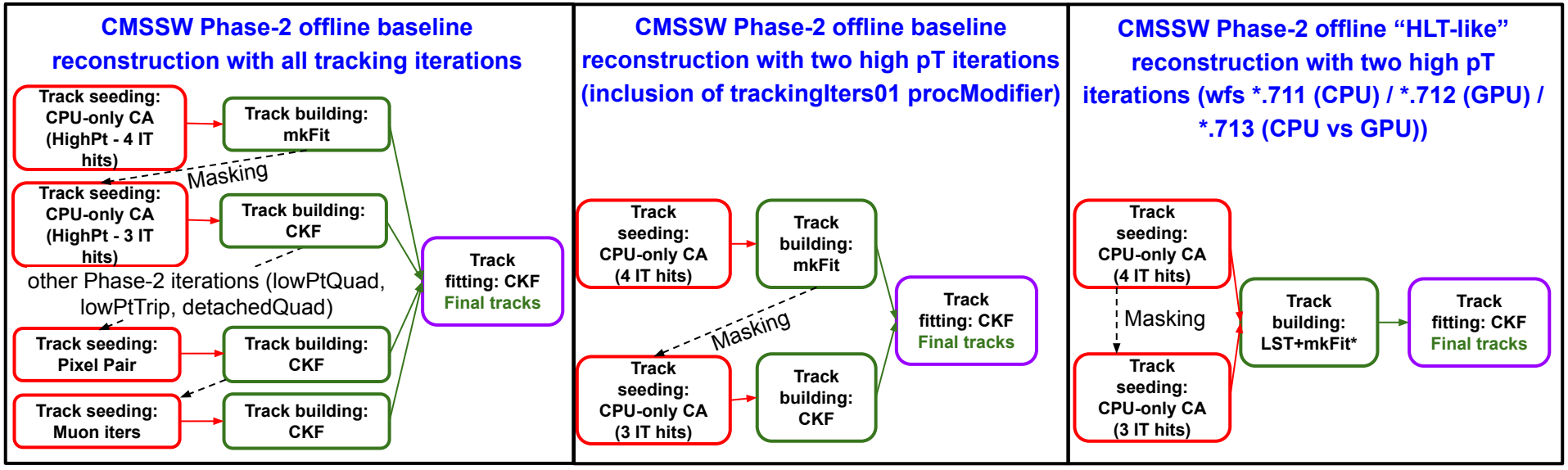


- Traditionally, track reconstruction consists of multiple iterations
  - Hits used in N^{th} iteration are masked for $(N+1)^{th}$ iteration
- **Metrics** for **physics performance** evaluation:
  - Efficiency: fraction of sim. tracks from signal (hard scattering) vertex associated to ≥ 1 reco'd track
  - Fake rate: fraction of misidentified reco'd tracks
  - Duplicate rate: fraction of reco'd tracks associated multiple times to the same SIM tracks
  - ➢ A reco'd track is associated to a sim. particle if >75% of its hits originated from this SIM particle
- Some more details on performance evaluation
  - Currently we use TrackingParticle information
    - With the MultiTrackValidator to study the performance of these objects
  - Trying to understand if we can easily switch to TruthGraph, and, if yes, how transparent it would be to the end user (in collaboration with Muon POG)

(Modified) Slide from Mario's presentation in CMS Week on 12 Jun 2026

Current Offline Reconstruction WFs

- Most of the algorithms in development are common online and offline
  - Currently, the focus is on the improvement of algorithms used for tracking at Phase-2 HLT, and to check the performance when used for offline reco
    - Same algorithms used online are planned to be exploited in first iterations of offline reconstruction
  - Tracking-only WFs that include most of online developments are available for offline phase-2 reconstruction of high pT prompt iterations starting from Inner Tracker (IT) quadruplets/triplets (only two iterations)
    - *.711 (CPU) / *.712 (GPU) / *.713 (CPU vs GPU)
    - Can be used for checking offline tracking performance



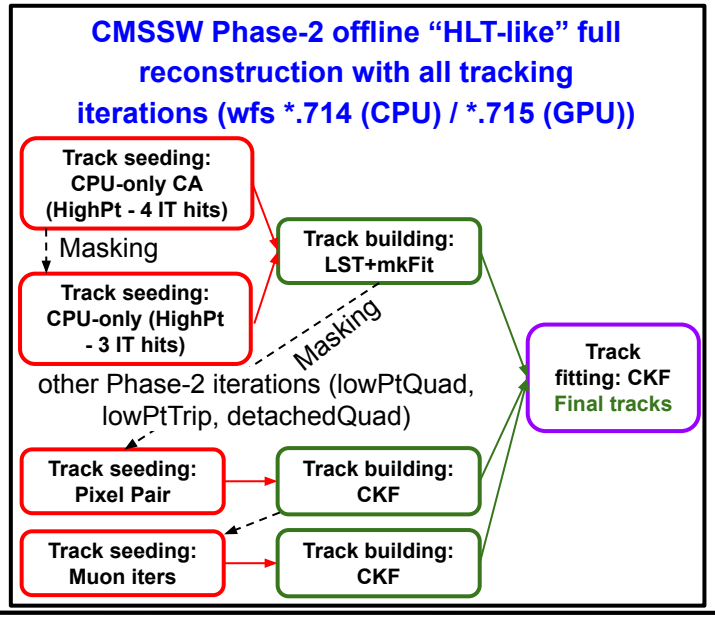
See Jan's presentation for a detailed description of the online Phase-2 algorithms

Quick Detour: Phase-2 HLT vs Offline

	HLT	Offline
Pixel seeds	Patatrack CA on the GPU, with OT extension	CPU-only CA quadruplets / triplets on the CPU
Beam spot	hltOnlineBeamSpot	offlineBeamSpot
LST geometry	hltLSTGeometry + hltESPModulesDevLST (0.8 GeV)	lstGeometryESProducer + lstModulesDevESProducer (0.6 GeV)
LST → seeds	OT hits dropped from pure pLS ending in the OT with ≤ 3 pixel hits (LSTOutputConverter.cc)	all hits kept
mkFit config	p_T^{min} 0.9 GeV; backward search kept if ≥ 2 pixel layers (1 if $\|d_0\| < 2$ cm)	p_T^{min} 0 GeV; no backward-search pixel requirement (defaults)
Candidate cuts	$p_T > 0.85$ GeV (0.8 for $\|\eta\| > 1.4$), ≥ 4 hits	none
Kalman fit	fitter–smoother for the LST step (χ^2 cut 20, ≤ 6 outliers; hltESPKFFittingSmootherForLSTStep)	FlexibleKFFittingSmoother (iterative-tracking default)
Selection	Torch DNN (WP 0.377, 0.267 displaced) → high purity	multiTrackSelector loose / tight / highPurity (χ^2/ndof, layers, d_0, d_z); displaced tracks pass
Merge	hltGeneralTracks: this collection only, $p_T > 0.8$ GeV	earlyGeneralTracks (5 iterations) + muon-seeded → generalTracks
Later iterations	single iteration	lowPtQuad, lowPtTriplet, detachedQuad, pixelPair
Muon seeded	muon L3 inside-out / outside-in in the muon paths, separate collections (e.g. hltIter0Phase2L3FromL1TkMuonCtfWithMaterialTracks)	muonSeededTracksInOut, muonSeededTracksOutIn, merged into generalTracks

Future Offline Reconstruction WFs

- How tracking will look like for phase-2?
  - It will include full RECO WFs that use online developments in the same first two high pT iterations, but keep the remaining iterations in tracking untouched (PR #52108)
  - It will include distinct algorithms:
    - Iterative heterogeneous CA for offline
      - Current reconstruction uses serial-only CA with IT-only hits
      - Use of OT vector hits in extended CA (see Marco/Jan presentations)
    - mkFit (detailed in Jan's presentation)
      - Since Run 3 mkFit is already used for track building (around 90% of tracks built offline)
        - For Phase-2 it was included in first iteration only
      - Include mkFit as track building algorithm for other iterations
    - mkFit fit (detailed in Jan's presentation)
      - Track fitting is currently performed by CKF
      - Already added in a non-default online workflow
      - Include it in offline reconstruction as well
    - DNN for tracks selection
      - It's already in online baseline
      - Use it for offline LST+mkFit configuration
  - Can the algorithms already be tested?
    - Yes, to some extent, but some still have to be included offline



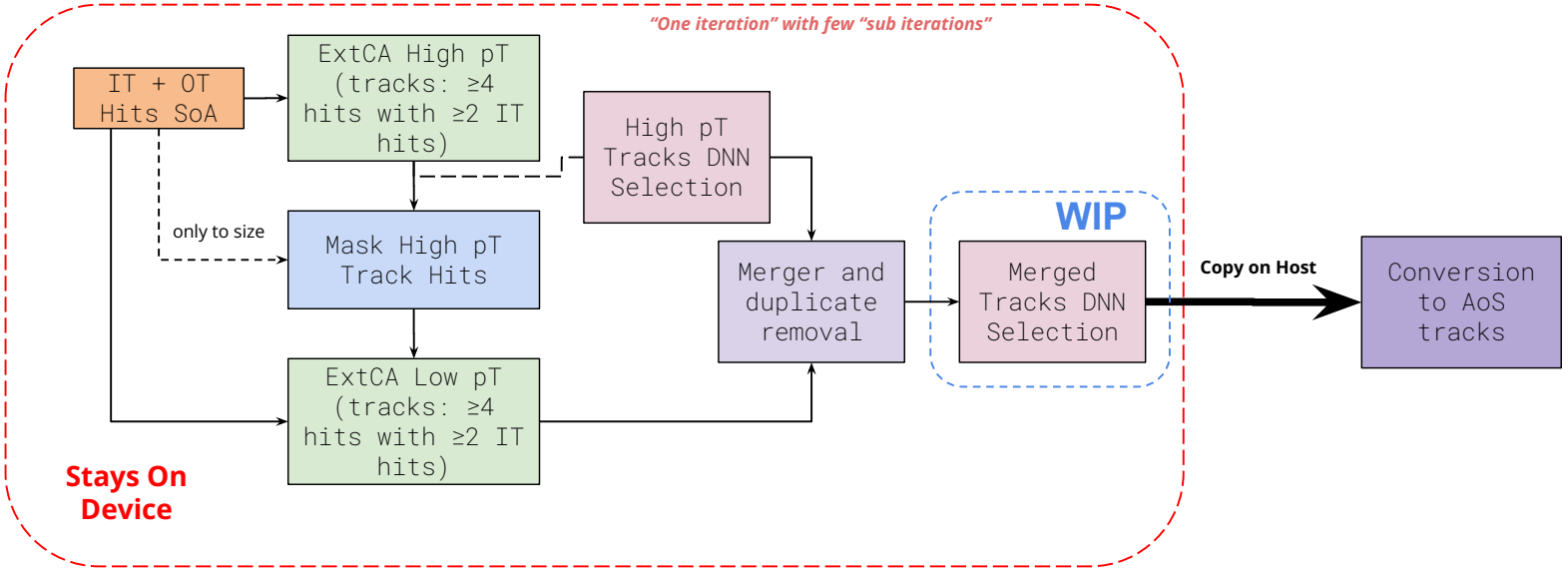
Future Iterations and Cosmics (and ML)

- How tracking will look like for phase-2?
  - In will include distinct iterations or improve current iterations
    - Tracking in dense environments
    - Muons
    - Given the changes in the seeding algorithm some of the existing iterations can be modified
  - New cosmics reconstruction
    - Phase-1 algorithm has hard-coded geometry and measurement types
    - Mainly for alignment

- What about ML for tracks reconstruction?
  - Transformers + Object Condensation for One-Shot Track Reconstruction using LST Triplets (T3s)
    - Model to embed LST T3s into a latent space using Object Condensation, pulling segments from the same track together and pushing others apart
    - Most recent developments:
      - Improved model and checked performance for segments with ≥ 4 layers (DP-2026-030)
  - Still under discussion

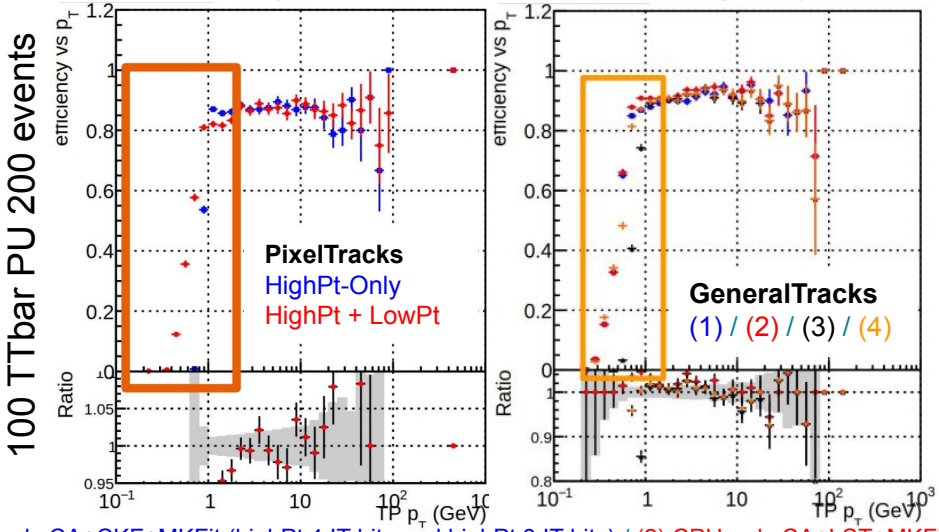
Future "Pixel" Tracking for Offline

- Internal hit-masking and multiple seeding iterations (B. Orzari, A. Di Florio; PR #51268)
  - Offline CA built for multiple interactions but not heterogeneous portability, such as online CA
    - First step towards Phase-2 offline reconstruction would be to include heterogeneous capability for seeding
  - Currently, online CA lacks the capability to accept the clusters (or hits) masking for multiple iterations
    - Removes used clusters (or hits) when building doublets which reduce combinatorics and make it easier to target different/harder regions
  - Timing/memory performance not studied in detail for the developments alone, due to lack of machines to study offline computing performance, but results from ITACA (L. Ferragina, NGT-3.1.1; **HLT-only**) can be checked to have an estimate of its performance
  - PR was opened in June and it needs to be merged for further developments (combination with other algorithms, dedicated tracking-only/full-RECO workflow)

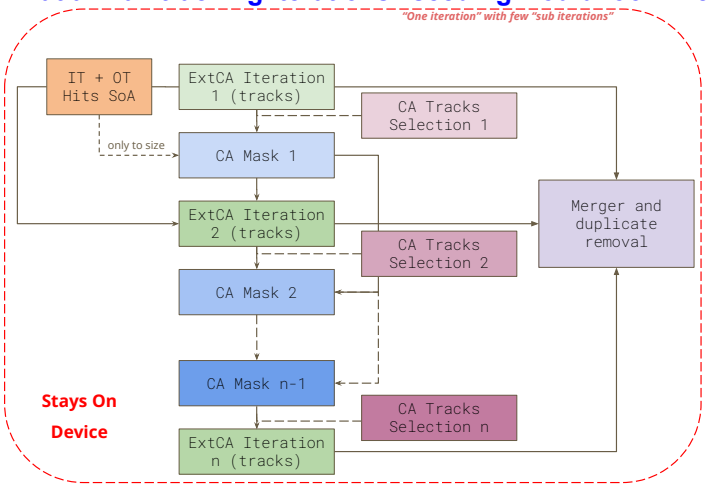


Future "Pixel" Tracking for Offline

- Internal hit-masking and multiple seeding iterations (B. Orzari, A. Di Florio; PR #51268)
  - Offline CA built for multiple interactions but not heterogeneous portability, such as online CA
    - First step towards Phase-2 offline reconstruction would be to include heterogeneous capability for seeding
  - Currently, online CA lacks the capability to accept the clusters (or hits) masking for multiple iterations
    - Removes used clusters (or hits) when building doublets which reduce combinatorics and make it easier to target different/harder regions
  - Timing/memory performance not studied in detail for the developments alone, due to lack of machines to study offline computing performance, but results from ITACA (L. Ferragina, NGT-3.1.1; **HLT-only**) can be checked to have an estimate of its performance
  - PR was opened in June and it needs to be merged for further developments (combination with other algorithms, dedicated tracking-only/full-RECO workflow)



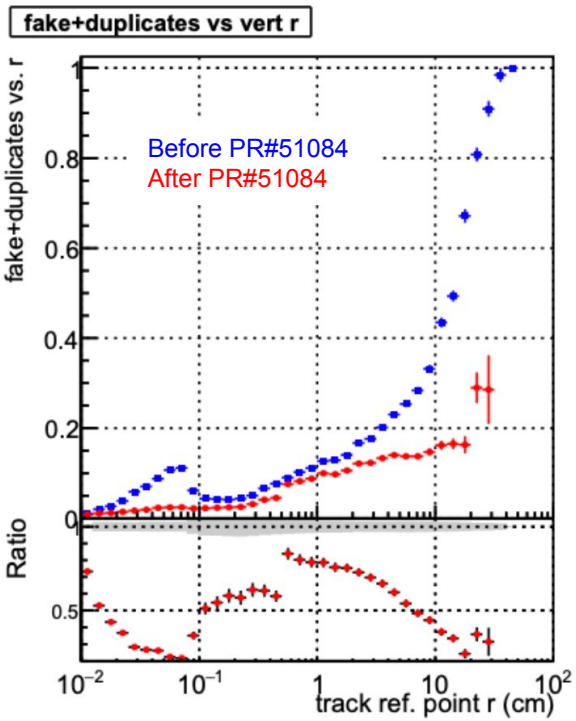
What an "all tracking iterations" seeding would look like



(1) CPU-only CA+CKF+MKFit (highPt 4 IT hits and highPt 3 IT hits) / (2) CPU-only CA+LST+MKFit (highPt 4 IT hits and highPt 3 IT hits) / (3) Masked CA+LST+MKFit (highPt 4 IT hits and highPt 3 IT hits) / (4) Masked CA+LST+MKFit (highPt 4 IT hits and lowPt 4 IT hits)

Future Track Selection for Offline

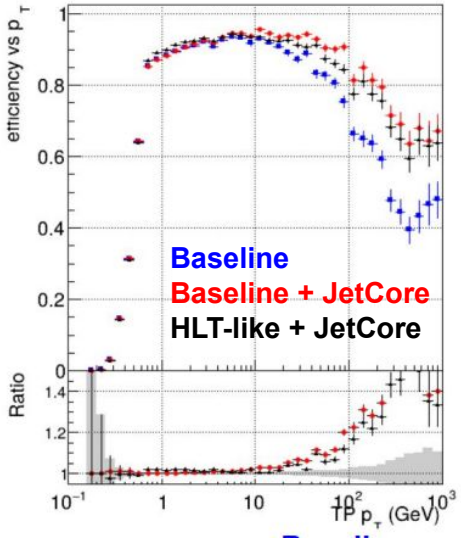
- ML Models to determine track quality
  - Similar to Run 3, develop a ML model to classify tracks based on track quality (e.g., “loose” vs. “high-purity” tracks); trained DNN to classify HLT full tracks; uses distinct working points for prompt and displaced tracks (A. Arora, J. Chismar; PR #51084)
    - Good physics performance achieved by maintaining efficiency while decreasing fake+duplicate rates
    - It’s already in online baseline, it’s coming for offline LST+mkFit configuration
      - Works as follows
        - Track classifier ≔ PyTorch DNN that scores every fitted track (genuine vs fake)
          - 15 features per track (box below); one score per track
          - Inference through Alpaka + PyTorch; CPU backend in the menu (alpaka_serial_sync)
        - Cut: keep if score ≥ 0.377, or if |dxy| > 0.5 cm and score ≥ 0.267
          - Displaced tracks get a looser cut; tuned for 99.5%
          - Same cuts for loose / tight / highPurity; only highPurity tracks are kept

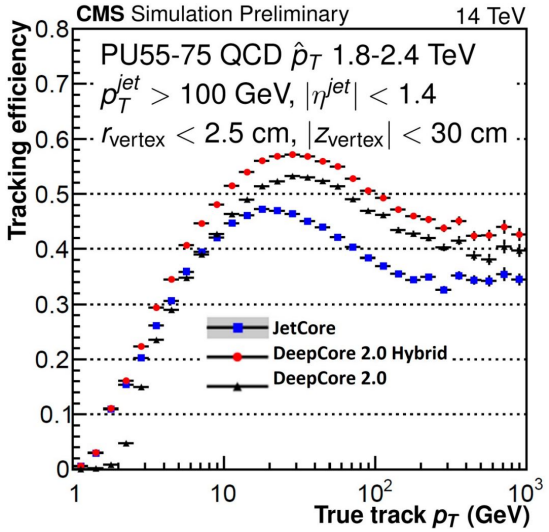


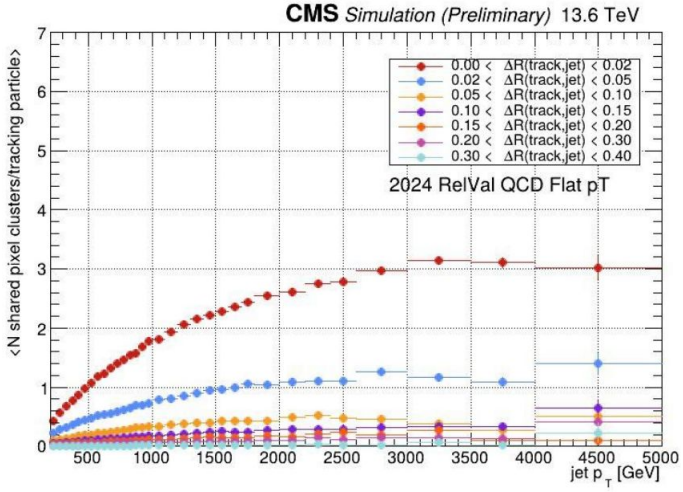
Input features:
d_{xy}, d_z to the beam spot + errors, η and φ + errors, normalised χ^2, ndof, lost inner / outer hits, layers without measurement, valid pixel hits, valid OT hits

Future Iterations for Offline

- Tracks reconstruction in dense environments: JetCore
  - Increase high pT tracks physics performance
  - Include JetCore in offline tracking reconstruction in combination with LST+mkFit
    - The JetCore iteration is currently not included for the phase-2 offline tracks reconstruction
      - Functionality is available, but not added as another iteration
      - Needs the inclusion of pre-splitting iteration
      - Additional pT cut requirements have to be used to reduce the timing of this iteration
    - Work is ongoing to include this offline (C. Hoang, M. Masciovecchio)
  - Recently found person-power to resume work on DeepCore
    - ML algorithm to help on defining track seeds in dense environments
  - Recent studies on the properties of clusters inside jets (C. Mauceri, L. Hay, M. LeBlanc)

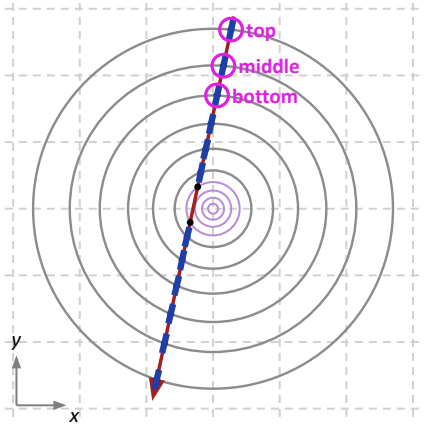






Future Cosmics Reconstruction (WIP)

- Grid cosmic seeder for Phase-2 cosmic tracking (M. E. Goblirsch-Kolb; PR #51586)
  - Space points are defined in a (x, y, z) grid
    - OT barrel: VectorHits (2 hits of a module ⇒ 1 point + direction) + leftover single hits; OT endcap: single hits; IT (pixel): all hits
  - Triplets are built from top to bottom in y
    - Hits in the same cell and the 3×3 cells one row below; different modules; ≤ 10 triplets per space point
  - Triplets cleaning
    - No zig-zag in z (> 1 cm) or x (> 2 cm); x–y slope compatible with the VectorHit direction (15%); curvature ⇒ pT ≳ 0.5 GeV
  - Seed fitting
    - Helix estimate, then Kalman fit through the 3 hits, both directions; pT > 1 GeV; ≤ 8 seeds per space point, ≤ 500 per event
  - Tracking: CKF + fit (to be optimised)
    - cosmicGridTripletSeeds ⇒ CKF (ckfTrackCandidatesP5) ⇒ fit (ctfWithMaterialTracksP5): Phase-1 cosmic Kalman chain, re-enabled as is, not yet tuned for Phase-2

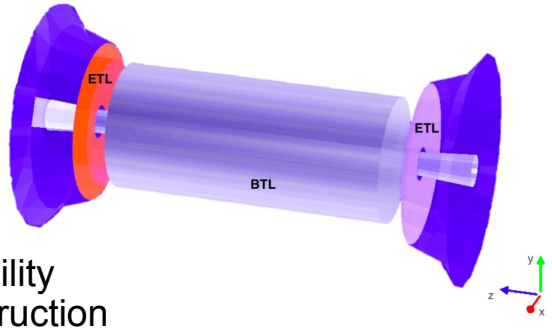


Transverse view: **cosmic muon**, ● space points, — VectorHit (barrel OT), ○ one triplet; dashed: grid cells (default config: one cell)

- A bit on the workflow related to cosmics reconstruction in Phase-2
  - Before: no viable Phase-2 cosmic reco; the Phase-1 cosmic seeder hard-codes Phase-1 geometry and measurement types ⇒ new geometry-agnostic seeder
  - Now: PR merged around one month ago; a private 500k-muon ALCARECO sample is already used for alignment studies
  - Next (ALCARECO discussion): the ALCARECO streams hard-code all Phase-1 cosmic track collections (CosmicTF, Regional, 0T, …), so today they run only if all point to ctfWithMaterialTracksP5 (not for production); APVMode crashes in Phase-2; tune CKF + fit for Phase-2

Tracking in 4D: MTD (brief) Overview

- MTD detector
  - Assign a time to the tracks of charged candidates
    - In high PU environment, will help to distinguish tracks/vertices that come from ROI and from PU
  - Divided into two subsystems
    - Barrel timing layer (BTL)
    - Endcap timing layer (ETL)

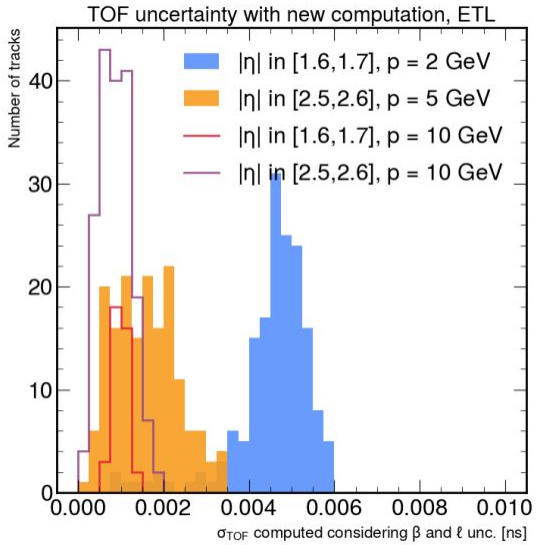


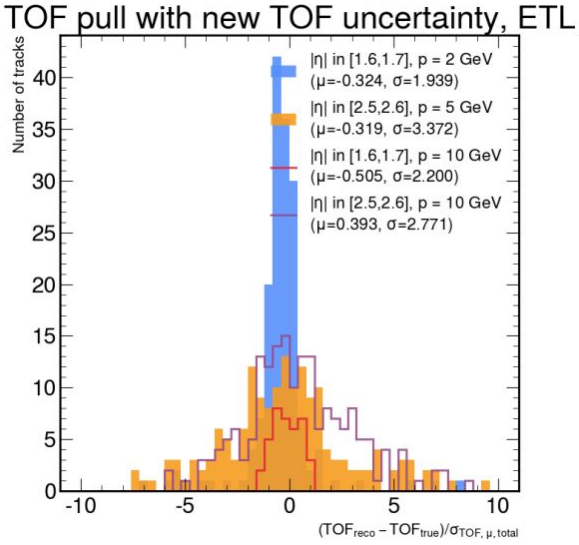
- MTD landscape
  - Brand “new” detector
    - Pros: no Run3 legacy to maintain/evolve/keep compatibility
    - Cons: the limited community is mostly focused on construction
      - Required software developments need some level of expertise not easily available among young students
        - Not trivially a matter of EPR, but of skills and readiness
- Tracking-related algorithms
  - MTD cluster - track matching and time propagation at the beam line
    - TrackExtenderWithMTD: association of tracker tracks to BTL/ETL clusters, track path length calculation and time back-propagation to the PCA to the beam line in different mass hypothesis;
      - TOFPIDProducer: remove mass hypothesis ambiguity using space and time compatibility with reconstructed primary vertices;
      - MTDTrackQualityMVAProducer: BDT evaluating a quality score for the track - MTD matching goodness
  - 4D Vertexing
    - 4D deterministic annealing
    - GNN for vertex building based on 4D information

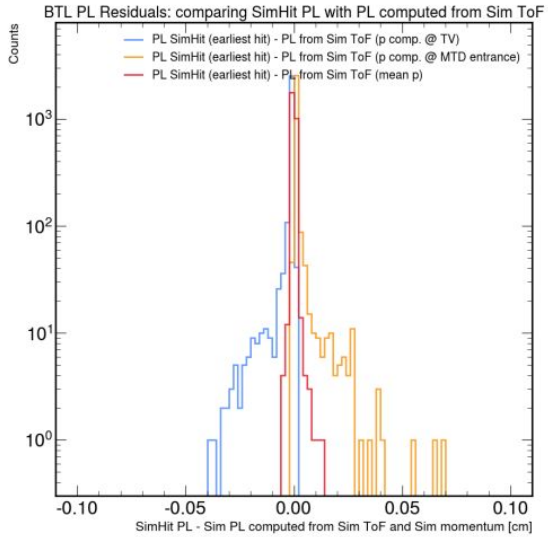
For more information see slides from Fabio’s presentation in Phase2 Software Days on 24 Sep 2026

Tracking in 4D: TrackExtenderWithMTD

- Extension of tracks with MTD information happens in 7 steps
  - Read tracker tracks -> Compute path length-> Match BTL / ETL hits -> Refit with MTD hits -> Back-propagate time -> PID likelihoods -> Write outputs
    - Outputs are: new track collection and 20 values related to the reconstruction performed in these steps (t_0, β, tofPi/K/P, match χ^2, MTD position, …) in the form of ValueMaps associated to generalTracks
      - Will be combined into a single object after some update concerning track - ETL matching
  - Some results (AN-2026/015)
    - ETL track TOF uncertainty under 6 ps
      - With narrow pulls
    - BTL track path length residuals under 0.5 mm



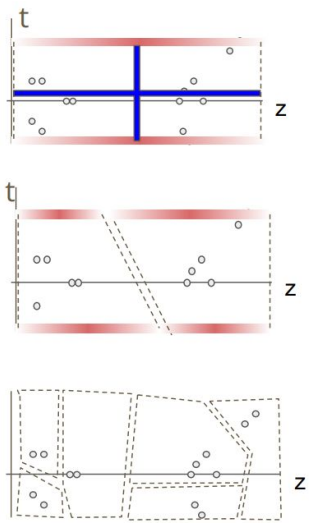


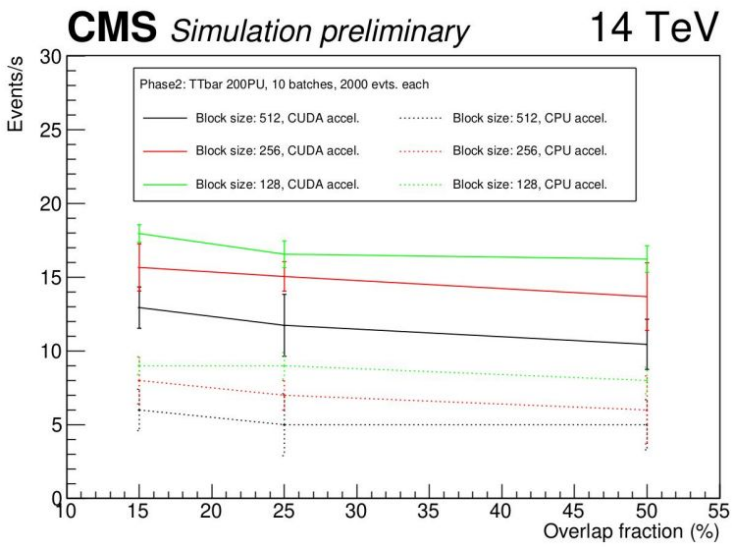


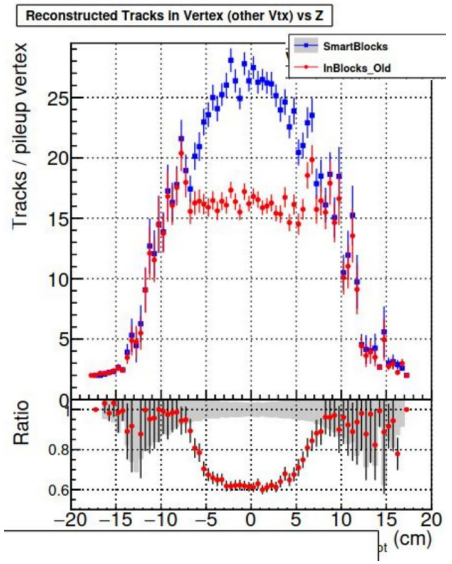
For more information see slides in dedicated MTD general / DPG Meeting on 19 Jun 2026

Future Vertexing for Offline

- What about vertices?
  - During Run 3 vertices were built with Deterministic Annealing algorithm and fitted by the adaptive vertex fitting algorithm
    - Both serial-only and too computationally expensive for high PU
- For Phase-2, some ideas being explored
  - Parallelized offline 3D and 4D vertexing reconstruction (C. Erice Cid; PR #50766 **only adds 3D**)
    - Introduces Deterministic Annealing in Blocks and weighted mean computation to fit vertex properties
      - Shows comparable/better performance in terms of physics with baseline DA; better throughput up to one order of magnitude
    - Currently on the stage of technical optimization of the algorithm and plans for smart block border assignment. Also, it touches on the subject of having common SoAs for some data formats which should be discussed with other developers

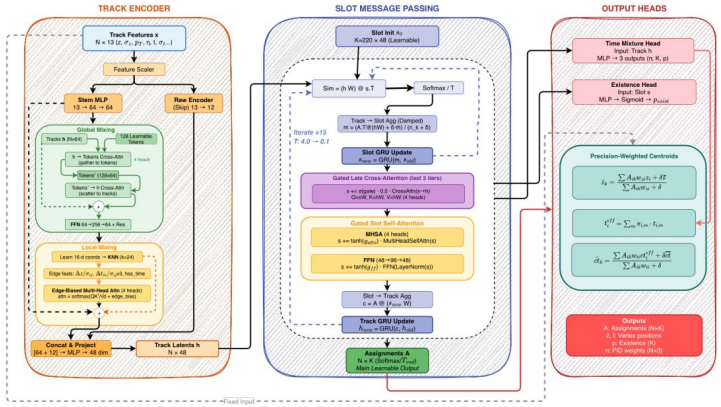


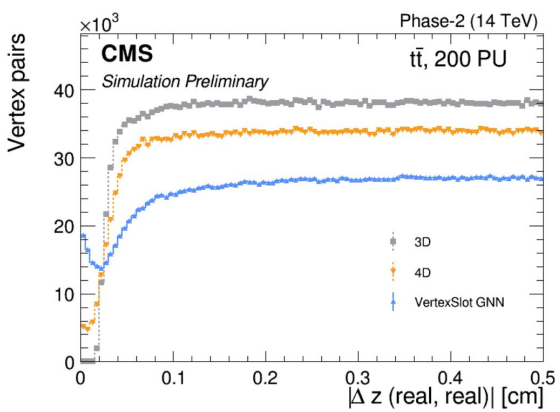


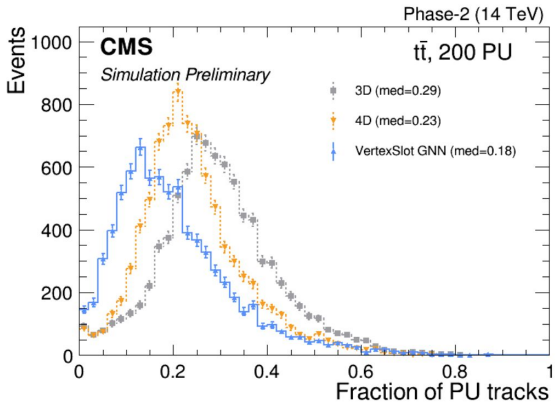


Future Vertexing for Offline

- What about vertices?
  - During Run 3 vertices were built with Deterministic Annealing algorithm and fitted by the adaptive vertex fitting algorithm
    - Both serial-only and too computationally expensive for high PU
- For Phase-2, some ideas being explored







  - Phase-2 primary-vertex reconstruction with the vertex-slot GNN (Alpaka inference), behind the vertexSlotGNN process modifier (DP-2026-042; PR #51862)
    - Vertex Slot Model: slots compete for tracks via soft attention; bidirectional message passing with per-event temperature annealing
      - Shows good physics performance in terms of reduced leading-vertex PU contamination, reconstructs comparable numbers of real vertices, and retains close vertex pairs that DA tends to merge
    - PR currently under review by the TRK POG and MTD DPG

More updates shown in Prabhat's presentation at MTD DPG meeting 9 Oct 2026

Some Topics for Discussion

- Detector material description in reconstruction
  - CMSSW limitations: material properties attached to sensitive surfaces (suboptimal description of passive regions); material when missing hits not accounted for
  - Possible solutions
    - Finer granularity description with CMSSW tools (see Nazar/Jan presentations)
    - Building of CMS geometry with ACTS, including passive material maps (L. Damenti; also see Nazar's presentation)
      - It is a long-term goal, but it also needs more person-power to continue studies for Phase-2
      - Can be a joint effort between TRK POG and TRK DPG
- Failure scenarios
  - It will be important to stress-test the Phase-2 algorithms on scenarios where parts of the detector are failing
    - Masked modules/sensors can be included with PR #52130
      - Would be interesting to have some samples for full sector failure scenarios to check the tracking performance
      - Would also be interesting to study how much failure-realistic scenarios implies on the need to re-train the track quality selectors/DNN for duplicate/fake rejections
- Automatic checks of performance
  - During CMS Week's Track 1: DQM and RelVal monitoring for online and offline
    - BPH asked to have a continuous central monitoring
      - Z(μμ), J/Psi (μμ, low pT), (K_S^0, Λ ?)
      - Tracking efficiency vs η/pT monitored
  - We would like to implement automatic checks for tracking performance while data taking is happening
    - Together with Muon POG collaboration if using Tag and Probe technique
- Many open or upcoming PRs
  - Reporting at HLT and RECO meeting
  - We keep an eye on PRs labelled as 

Some Topics for Discussion

- DQM plots cleaning
  - Also during CMS Week's Track 1: DQM and RelVal monitoring for online and offline
    - DQM cleanup for tracking POG (PR #52203)
- Phase-2 data formats
  - Need to decide the data formats for phase-2 (MiniAOD, NanoAOD)
    - Discussed during BPH Workshop and CMS Analysis Data Formats Developments
      - Needs person power
- Tracking within other POGs
  - E/γ
    - Pixel track seeds are reconstructed to match energy deposits in ECAL super clusters
    - Most recent developments:
      - Algorithm to perform SC-seed matching in parallel with GPUs, and contact with TRK POG for usage of already existing doublet and triplet producer modules to create trajectory seeds (C. Wessel, C. Koraka)
  - Muon
    - Optimizing inside-out reconstruction using single iteration with DNN quality selection (see Celia's presentation at Phase-2 Software Days - 25 Sep 2026)
    - Using distinct information from tracks to improve reconstruction in ME0 region
    - Speed up tracker muon reco with ML-based cleaning
  - JetMET
    - In contact to not overlap/diverge with vertexing and tracking developments
    - Recent contribution at TRK POG meeting to discuss about regional tracking
      - Track building with priority to direction of jets with highest energy
  - Tau
    - Also recent contribution at TRK POG meeting to discuss about truth graph usage, timing information for Phase-2 Taus and ongoing projects

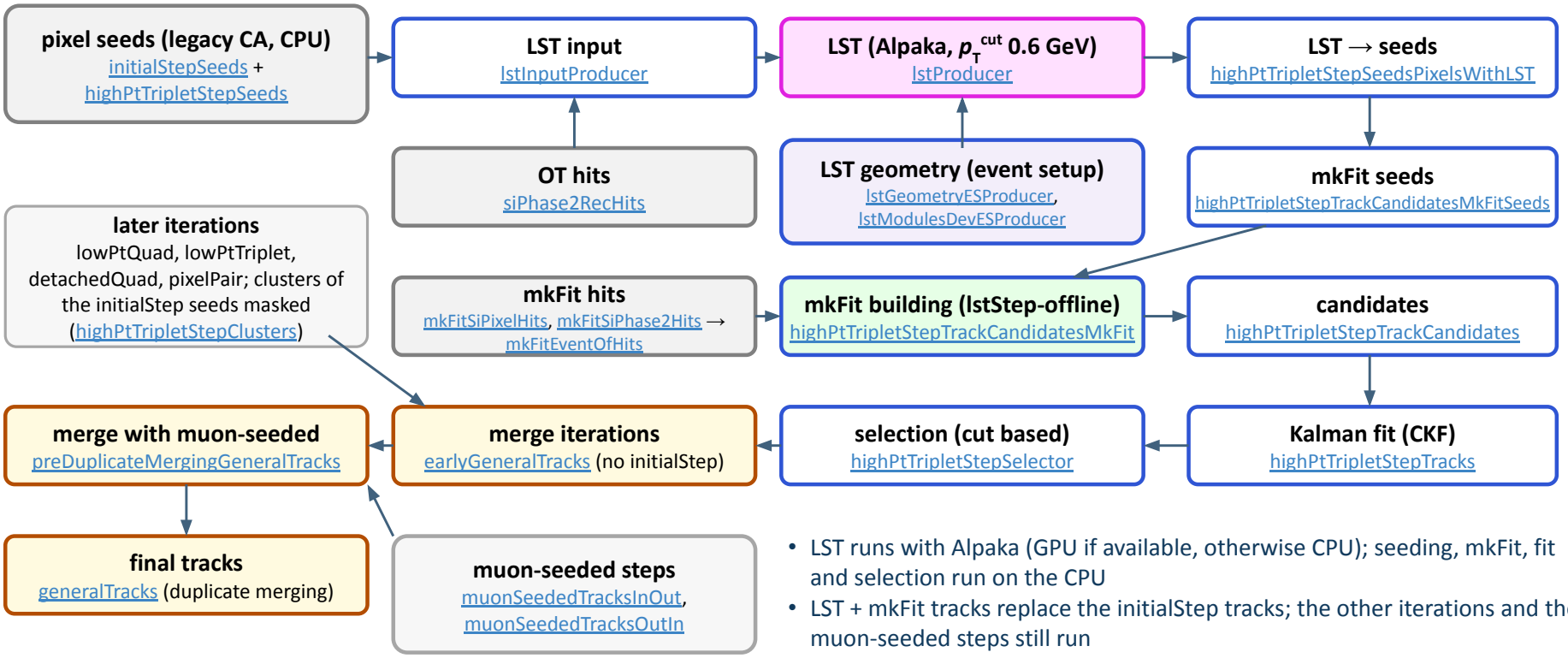
Summary

- Current offline tracking already provides the possibility to test some of the online algorithms in offline reconstruction
  - WFs *.711 - *.715 allows for CPU-only CA+LST+mkFit test in offline reconstruction
- Future offline tracking is evolving towards a more performant workflow both in terms of physics and throughput
  - Seeding
    - Masked heterogeneous maintains speed seen online while offering the flexibility of multiple iterations offline
    - LST seeding using OT hits allows for prompt and displaced seeds to be reconstructed from the beginning of track building
  - Building/Fitting
    - mkFit building and fitting speeds up tracking while maintaining or improving physics performance
  - Selection
    - ML models reduce fake+duplicate rate while maintaining efficiency
  - New iterations with respect to baseline
    - JetCore iteration increases physics performance for high pT tracks
  - Vertexing
    - Explore the parallelization of DA and new vertex fitting for higher throughput while preserving/improving physics performance
    - Utilize ML methods for 4D vertex reconstruction
- Open topics
  - Detector material description for online and offline reconstruction
  - Inclusion of failure scenarios samples
  - Automatic validation for continuous central monitoring
  - Decide on phase-2 DataFormats

Backup

Phase-2 Offline Reconstruction Chain

highPtTripletStep iteration (Phase-2, LST + mkFit modifiers)



- LST runs with Alpaka (GPU if available, otherwise CPU); seeding, mkFit, fit and selection run on the CPU
- LST + mkFit tracks replace the initialStep tracks; the other iterations and the muon-seeded steps still run

cms-sw/cmssw@f6063aa: HighPtTripletStep, RecoTracker/LST/python, preDuplicateMergingGeneralTracks

Phase-2 Offline Collections

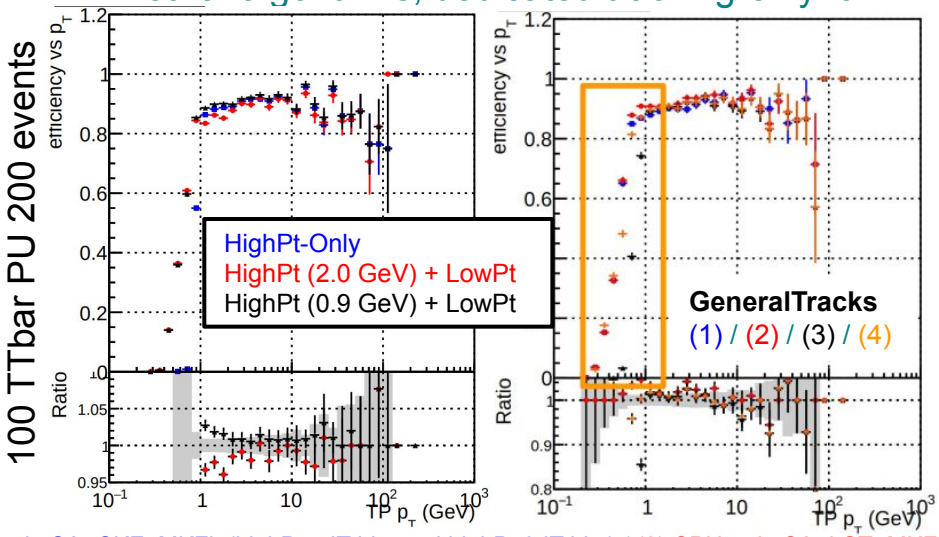
	HLT (HLT_75e33, default)	Offline (LST + mkFit process modifiers)
OT hits	hltSiPhase2RecHits	siPhase2RecHits
Pixel seeds	hltInitialStepSeeds (Alpaka pixel tracks)	initialStepSeeds + highPtTripletStepSeeds
LST input	hltInputLST (seeds p_T > 0.8 GeV − 2σ)	lstInputProducer (seeds p_T > 0.1 GeV − 2σ)
LST	hltLST (p_T^{cut} = 0.8 GeV)	lstProducer (p_T^{cut} = 0.6 GeV)
LST → seeds	hltInitialStepTrajectorySeedsLST	highPtTripletStepSeedsPixelsWithLST
mkFit hits	hltMkFitSiPixelHits, hltMkFitSiPhase2Hits → hltMkFitEventOfHits	mkFitSiPixelHits, mkFitSiPhase2Hits → mkFitEventOfHits
mkFit seeds	hltInitialStepMkFitSeeds	highPtTripletStepTrackCandidatesMkFitSeeds
mkFit building	hltInitialStepTrackCandidatesMkFit (lstStep.json, p_T^{min} 0.9)	highPtTripletStepTrackCandidatesMkFit (lstStep-offline.json, p_T^{min} 0)
Candidates	hltInitialStepTrackCandidates (p_T > 0.85 GeV, ≥ 4 hits)	highPtTripletStepTrackCandidates (no candidate cuts)
Kalman fit (CKF)	hltInitialStepTracks	highPtTripletStepTracks
Selection	Torch DNN → hltInitialStepTrackSelectionHighPurity	highPtTripletStepSelector (cut based, displaced pass-through)
Merged tracks	hltGeneralTracks	earlyGeneralTracks (no initialStep) + muon-seeded → preDuplicateMergingGeneralTracks → generalTracks

Same algorithms, different wiring

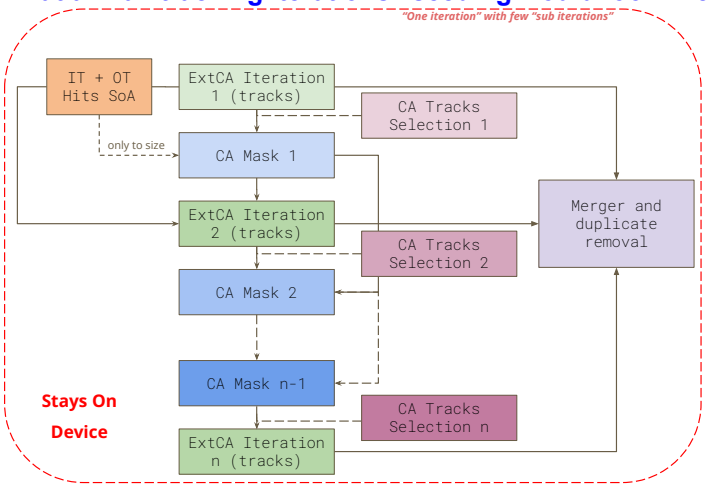
- HLT: LST + mkFit form the **initialStep**; offline they run inside the **highPtTripletStep** iteration and the initialStep tracks are not merged (HighPtTripletStep)
- Offline modifiers: trackingLST, seedingLST, trackingMkFitHighPtTripletStep (Phase-2). Offline uses lower thresholds (LST 0.6 GeV, mkFit p_T^{min} 0, no candidate cuts) and a cut-based selection instead of the DNN

Future "Pixel" Tracking for Offline

- Internal hit-masking and multiple seeding iterations (B. Orzari, A. Di Florio; PR #51268)
  - Offline CA built for multiple interactions but not heterogeneous portability, such as online CA
    - First step towards Phase-2 offline reconstruction would be to include heterogeneous capability for seeding
  - Currently, online CA lacks the capability to accept the clusters (or hits) masking for multiple iterations
    - Removes used clusters (or hits) when building doublets which reduce combinatorics and make it easier to target different/harder regions
  - Timing/memory performance not studied in detail for the developments alone, due to lack of machines to study offline computing performance, but results from ITACA (L. Ferragina, NGT-3.1.1; **HLT-only**) can be checked to have an estimate of its performance
  - PR was opened in June and it needs to be merged for further developments (combination with other algorithms, dedicated tracking-only/full-RECO workflow)



What an "all tracking iterations" seeding would look like

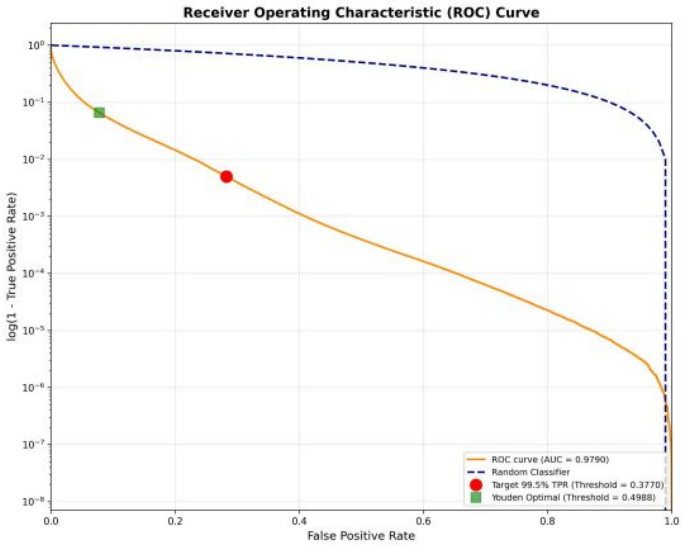


(1) CPU-only CA+CKF+MKFit (highPt 4 IT hits and highPt 3 IT hits) / (2) CPU-only CA+LST+MKFit (highPt 4 IT hits and highPt 3 IT hits) / (3) Masked CA+LST+MKFit (highPt 4 IT hits and highPt 3 IT hits) / (4) Masked CA+LST+MKFit (highPt 4 IT hits and lowPt 4 IT hits)

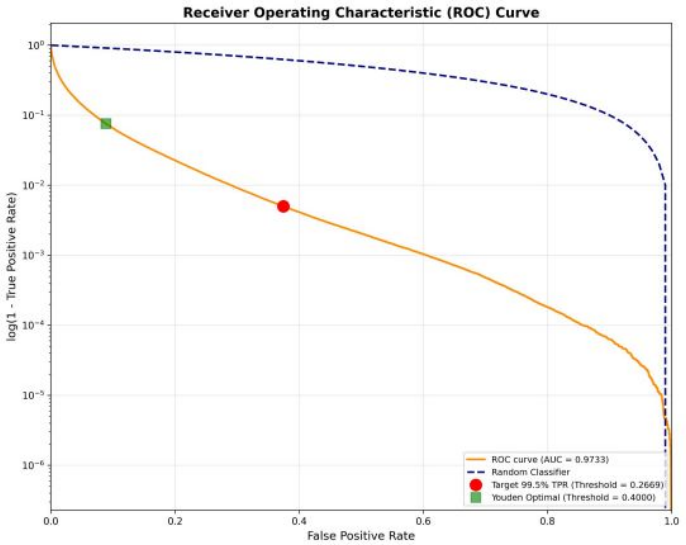
DNN for HLT Performance

DNN for HLT Track Purity in Run-4 - Performance

- Working point chosen based on a target recall of 99.5%
- For displaced tracks, a separate working point based on a target recall of 99.5% of displaced tracks only is used



All tracks



Displaced tracks

Slide from Aashay and Jade's presentation in Machine Learning Applications in Reconstruction - 4 Jun 2026

Cosmics Reco Phase-2

Summary of efficiency loss

State of the checks so far:

- Main problem seems to be collecting all the hits on the true trajectory
  - they (visually - event display) align perfectly with the seed trajectory
- Found cases where the navigation and detector module lookup missed hits from cosmics
  - Experimentally: Seems to mainly matter for non-vertical tracks, a third of the total loss
- Two thirds of the additional track loss w.r.t Phase-1 seem to be related to trajectories where just 4 out of 5 required hits are found
  - Additional hits being missed for vertical tracks
  - Some are **really** difficult trajectories violating the "extend seed in one direction" paradigm
  - Can test a brute-force extension mechanic in "empty" cosmic events for fun :)

DESY. 39

Slide from Maximilian's presentation in Tracker DPG / Tracking POG - 12 Ago 2026

Cosmics Reco Phase-2

More fixes for Phase-2

Found by Sandra: **APVMode** for cosmics **crashes the run** in Phase-2 - need to comment for successful processing

```
----- Begin Fatal Exception 28-Aug-2026 10:35:14 CEST-----------------------
An exception of category 'NoRecord' occurred while
   [0] Processing  stream begin Run run: 1 stream: 0
   [1] Calling method for module APVModeFilter/'apvModeFilter'
Exception Message:
No "SiStripLatencyRcd" record found in the EventSetup.

 Please add an ESSource or ESProducer that delivers such a record.
----- End Fatal Exception -------------------------------------------------
```

Seems to look for a Si strip latency record? Will this concept return for Phase-2, or should we remove it?

```
# Ensure the correct APV mode for cosmics
# -----------------------------------------------------------------------------
if collection in ("ALCARECOTkAlCosmicsCTF0T",
                  "ALCARECOTkAlCosmicsInCollisions"):
    process.load("Alignment.CommonAlignment.apvModeFilter_cfi")
    process.apvModeFilter.apvMode = "deco" if cosmics_deco_mode else "peak"
    from . import helper
    helper.add_filter(process, process.apvModeFilter)
```

```
47  # Ensure the correct APV mode for cosmics
48  # -----------------------------------------------------------------------------
49+ # if collection in ("ALCARECOTkAlCosmicsCTF0T",
50+ #                   "ALCARECOTkAlCosmicsInCollisions"):
51+ #     process.load("Alignment.CommonAlignment.apvModeFilter_cfi")
52+ #     process.apvModeFilter.apvMode = "deco" if cosmics_deco_mode else "peak"
53+ #     from . import helper
54+ #     helper.add_filter(process, process.apvModeFilter)
```

DESY. 8

Slide from Maximilian's presentation in Tracker Alignment - 6 Oct 2026

3D DA in Blocks Common SoA

Alpaka 3D code - Track/Vertex (common?) SoA

```
GENERATE_SOA_LAYOUT(TrackForVertexSoALayout,
        // columns: one value per element
        SOA_COLUMN(float, dxy2),
        SOA_COLUMN(float, dxy2AtIP),
        SOA_COLUMN(float, dz2),
        SOA_COLUMN(float, oneoverdz2),
        SOA_COLUMN(float, weight),
        SOA_COLUMN(float, sum_Z),
        SOA_COLUMN(int, kmin),
        SOA_COLUMN(int, kmax),
        SOA_COLUMN(bool, isGood),
        SOA_COLUMN(int, order),
        SOA_COLUMN(int, ttIndex),

        SOA_COLUMN(float, x),
        SOA_COLUMN(float, y),
        SOA_COLUMN(float, z),

        SOA_COLUMN(float, xAtIP),
        SOA_COLUMN(float, yAtIP),

        SOA_COLUMN(float, dx),
        SOA_COLUMN(float, dy),
        SOA_COLUMN(float, dz),

        SOA_COLUMN(float, dxError),
        SOA_COLUMN(float, dyError),
        SOA_COLUMN(float, dzError),

        SOA_COLUMN(float, px),
        SOA_COLUMN(float, py),
        SOA_COLUMN(float, pz),

        SOA_COLUMN(float, aux1),
        SOA_COLUMN(float, aux2),

        // Track-vertex association
        SOA_EIGEN_COLUMN(TrackToVertex, vert_sw),
        SOA_EIGEN_COLUMN(TrackToVertex, vert_se),
        SOA_EIGEN_COLUMN(TrackToVertex, vert_swz),
        SOA_EIGEN_COLUMN(TrackToVertex, vert_swE),

        // scalars: one value for the whole structure
        SOA_SCALAR(int32_t, nT),
        SOA_SCALAR(float, totweight))
```

Possible common dataformat with other SoA-based tracking efforts somewhat tricky to treat because different algorithms have different needs.

I.e. to the left is an screenshot of what we currently have on our PR.

- **Blue** are things that do not need to leave the clustering
- **Red** are things that we need as input/output.

Even **red** is quite different from what we have in other modules right now:

- I.e. TrackSoA (for pixel vertex reco) is much lighter.
- ZVertexSoA (for HLT vertex) is probably possible to match once we get rid of transient info in the vertex dataformat

Our plan is to provide a concrete proposal in the next few months but this will need to be discussed here (I think) as it touches several other groups <= Feedback welcome!

7

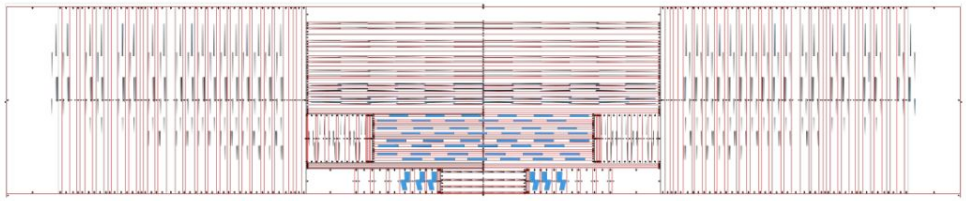
Slide from Carlos' presentation in TRK POG 21 Sep 2026

Material Modeling Using ACTS

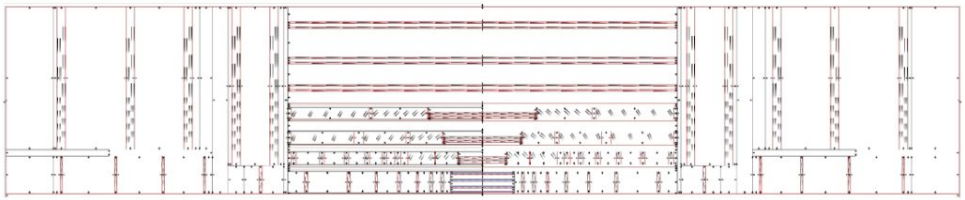
CMS Geometry and Material Mapping

- **Step I:** Build the CMS geometry using ACTS tools
- Basic principle → Layers arranged into volumes following a hierarchical structure
- Both Phase-1 and Phase-2 geometries are implemented

Phase-1 geometry:



Phase-2 geometry:



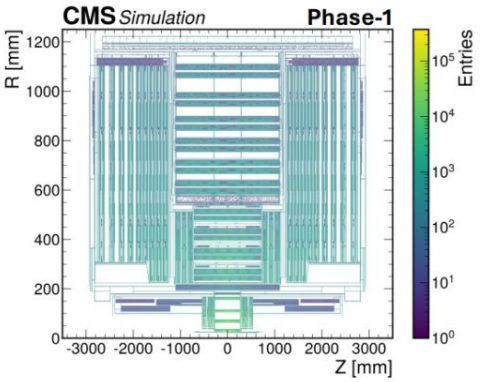
- Material layers (highlighted in red) have a configurable binned structure and can be flexibly placed within the geometry
- **Optimization:** (I) Number and position of material layers, (2) Number of bins

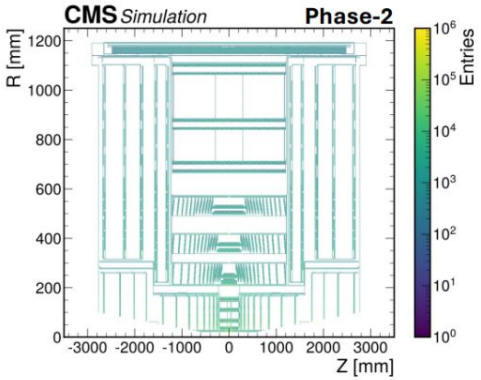
Slide from Lorenzo's presentation in Weekly Tracker DPG Meetings - Geometry & material budget - 25 Sep 2026

Material Modeling Using ACTS

CMS Geometry and Material Mapping

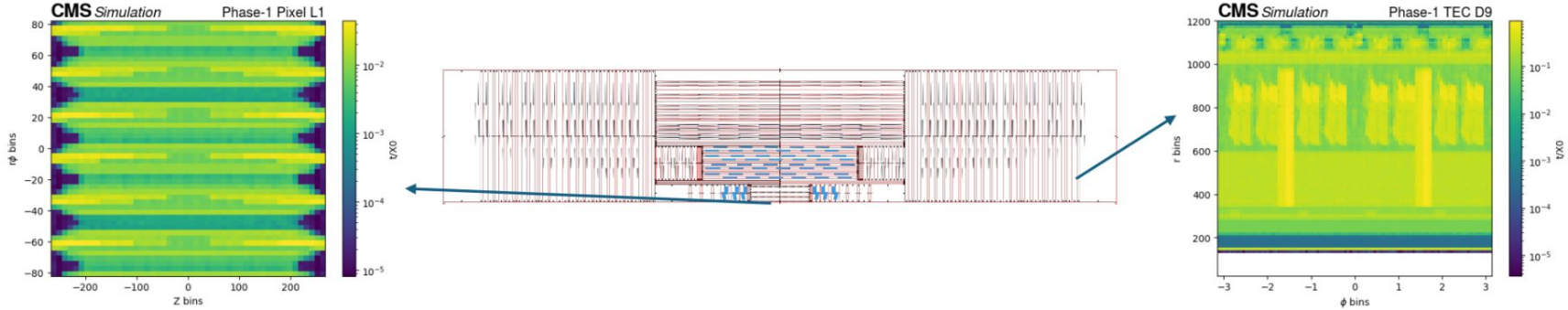
- **Step II:** Map passive material properties on material layers
- Basic principle → Probe passive material using geantinos propagating through the detailed Geant4 geometry description





- For each particle, material properties are assigned to the nearest bin of the nearest material layer
- Final maps are obtained by averaging the values assigned to each bin

- Examples:



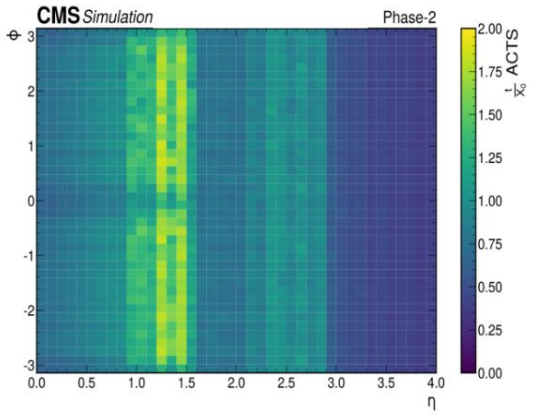
- $t =$ particle path length across material; $X_0 =$ Radiation Length

Slide from Lorenzo's presentation in Weekly Tracker DPG Meetings - Geometry & material budget - 25 Sep 2026

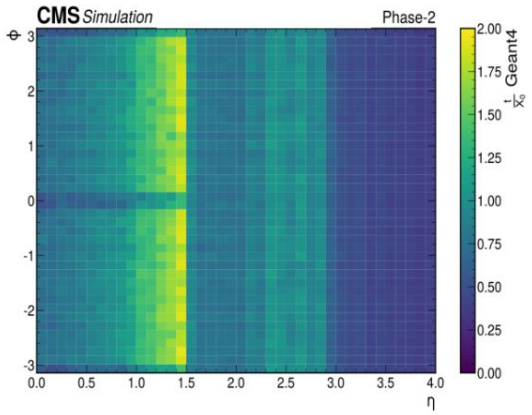
Material Modeling Using ACTS

Material Maps Validation

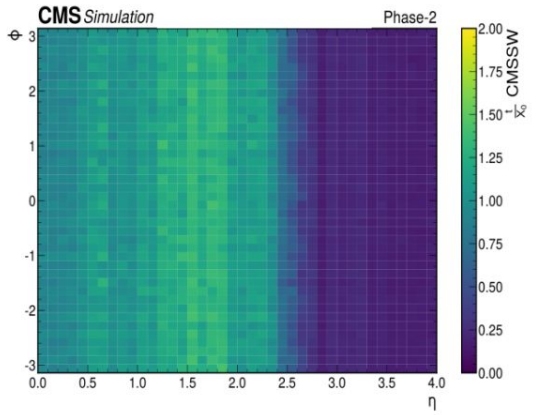
- **First test:** Use the ACTS-based propagator to validate the material maps
- Workflow: perform track refits on simulated 100 GeV muons sampling material properties at each propagation step
- **Results:**
  1. ACTS captures finer material details (better agreement with Geant4), while CMSSW appears more uniform
  2. No ϕ sub-structures in CMSSW (by construction)
- True for both Phase-1 and Phase-2 geometries



ACTS



Geant4



CMSSW

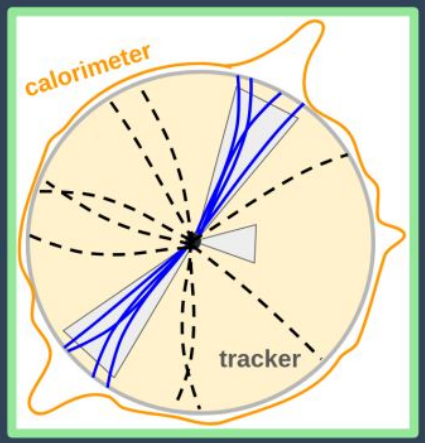
Slide from Lorenzo's presentation in Weekly Tracker DPG Meetings - Geometry & material budget - 25 Sep 2026

JetMET - Regional Tracking

Intro: Basics of Regional Tracking

Regional reconstruction:

- Track building with priority to direction of jets with highest energy
- For decision, need object that can be reconstructed rather fast:
  → could do selection based on **Calo jets** or even L1 seed jets



How to go about this:

- Option 1: build full-tracks only in region of interest, keep all pixel seeds
- Option 2: keep also pixel seeds only in interesting regions
- Option 3: *... many directions to be explored ...*

Previous developments along these lines:

- Historically: jet-plus-tracks algorithm during Run1 (basically, correct calo jet with momentum info from tracks), see e.g. CMS PAS JME-09-002 or JME-14-005
- Regions-of-interest (ROI) as mentioned e.g. in a paper (M. Tosi, 2014) before Run2 to reduce CPU-time
- Existing studies by BTV from before Run3 (see S. Wuchterl, 2021) on using ROIs around jets

29.09.2026 Bettina Lehtelä (Helsinki Institute of Physics) 8

Slide from Bettina's presentation in TRK POG - 29 Sep 2026